[HiNAS Control Standard Onboard Test Procedure]

Avikus

Accept by:
2026.05.22

Tested by: **Seunghyun Woo**

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Date : 2026.05.21

[Revision History]

Documents Number.	Revision Number.	Revision Date	Description
AVK-NAS-CTRLSTD-0040	1.0.0	2025-03-30	Draft
AVK-NAS-CTRLSTD-0040	1.0.1	2025-03-30	Added expected result
AVK-NAS-CTRLSTD-0040	1.0.2	2025-11-20	3.4 Autopilot fail test revised

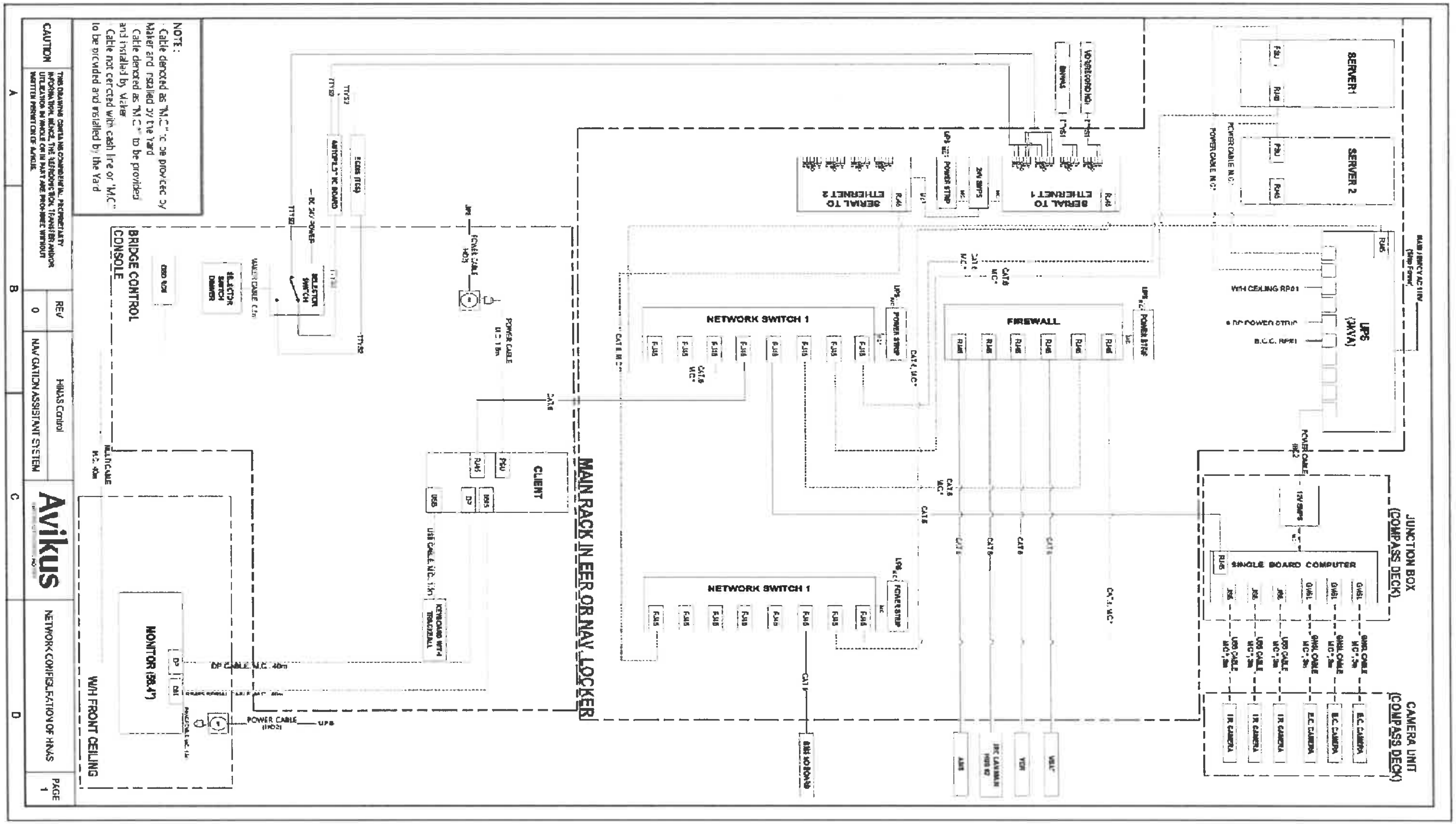
[Operator Information]

Operator	Position	Name	Signature
-	manager	Seunghyun Woo	

[Caution]

Contents

1. Wiring Diagram (Should be changed as per the ship specifications)



2. Check list for Testing

2.1 Performance Tests

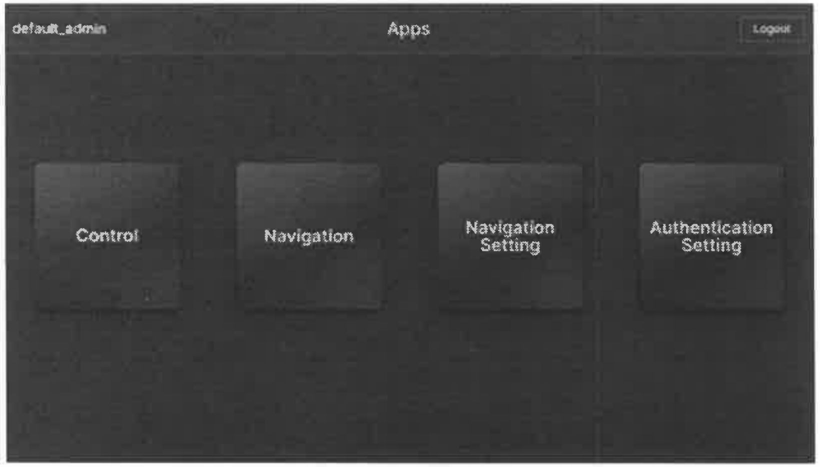
No	Article	Contents	No	Article	Contents
1	-	System health check test	6	IEC62065 -	BMS fail test
2	-	Basic functionality test	7	IEC62065 5.1.3.1	Power fail test
3	-	Route planning test	8	-	Operation mode change test – Mode selection unit
4	IEC62065 5.1.1.14 5.5.1	Autopilot fail test	9	-	Operation mode change test – SW Button
5	IEC62065 5.1.3.2 5.1.3.4 5.5.2~5.5.4	Sensor interfaces fail test			

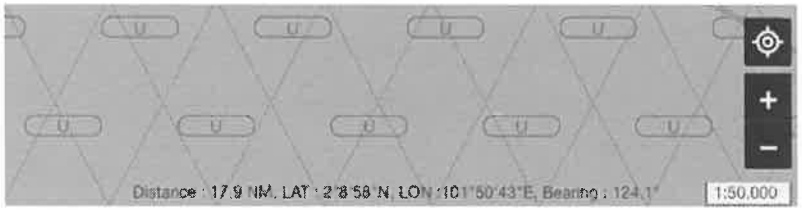
3. Test Contents & Procedure

3.1 System Health Check Test

Relevant Standard	Standard No.		
	Article.		
Purpose	The purpose of this test is to check the hardware equipment before actual test		
Procedure of test		Pass	Fail
1: Visual inspection items below. - Steering unit (Auto pilot) is installed on ship - Speed control unit (BMS) is installed on ship - Route sender (ECDIS) is installed on ship - HiNAS Control Standard is installed on ship		v	
Comment / Recommended			

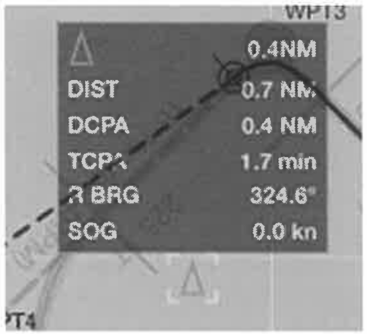
3.2 Basic Functionality Test

<table>
<tr><td rowspan="2">Relevant Standard</td><td colspan="3">Standard No. (MSC Circ.982)</td></tr>
<tr><td colspan="3">Article.</td></tr>
<tr><td>Purpose</td><td colspan="3">The purpose of this test is to evaluate display function of HiNAS Control Standard in normal condition performance</td></tr>
<tr><td colspan="2">Procedure of test</td><td>Pass</td><td>Fail</td></tr>
<tr><td colspan="2">1. Execute the HiNAS Control Standard product in apps page


2. Test zoom in/out function and see if map display scale is changed as per the control


3. Test display mode change function and see if display mode (Day, Night)/ (map only, camera only, both) / (Brightness) is changed as per the control</td><td>v</td><td></td></tr>
</table>

* Note: Camera and map view may vary with external environment

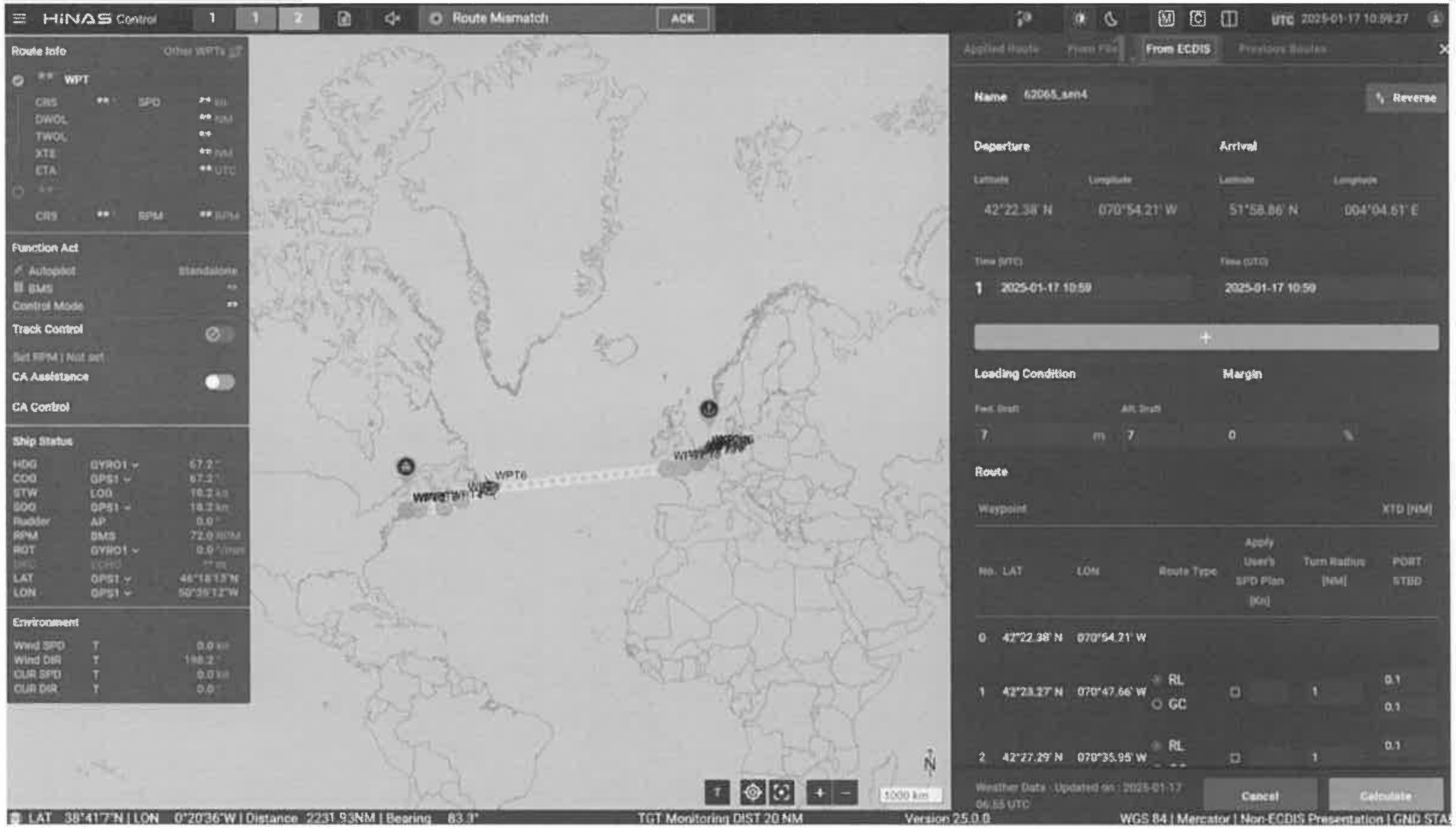


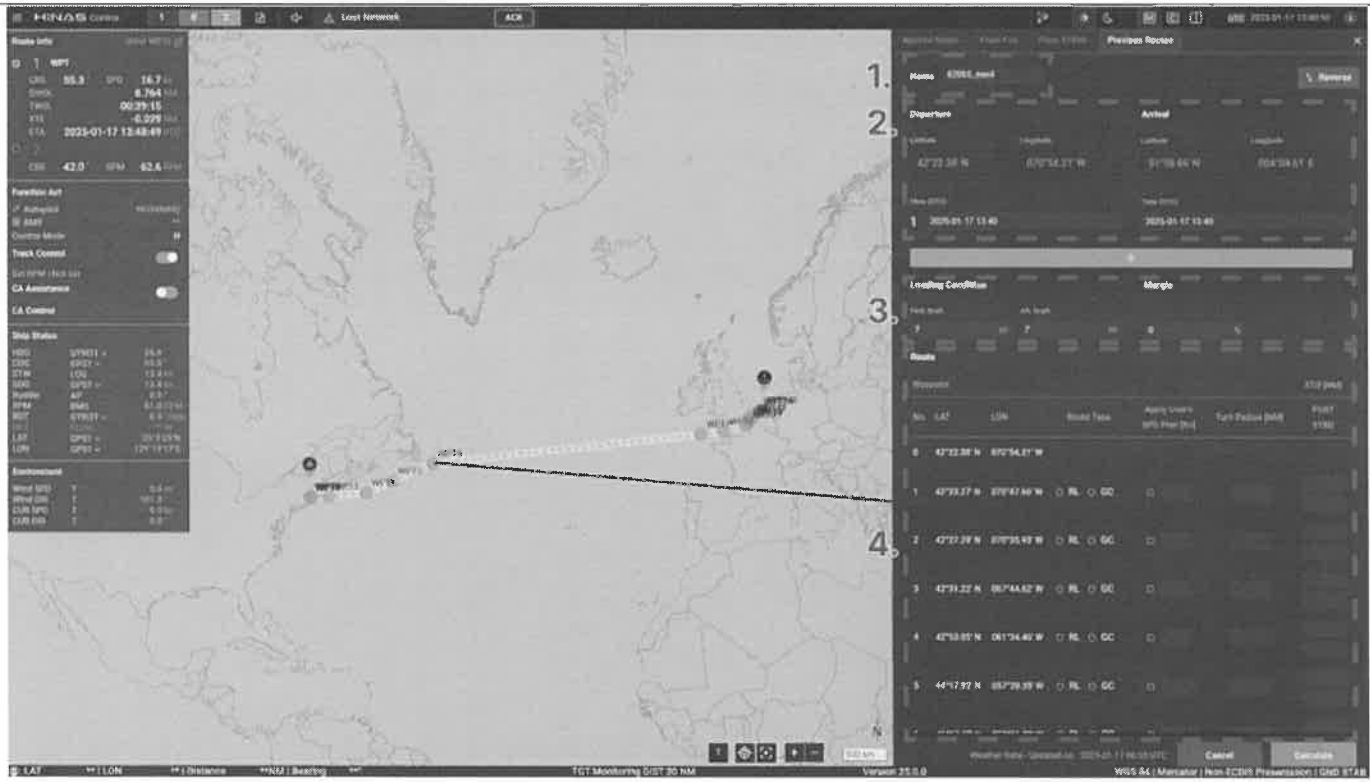
* Note: Target information may vary with external environment

Comment / Recommended

3.3 Route Planning Test

Relevant Standard			
Purpose	The purpose of this test to evaluate route planning API performance (no weather condition) and to evaluate route planning API interface performance		
Procedure of test		Pass	Fail
1. Plan any dummy route in ECDIS and activate route monitoring function of ECDIS 2. Open route planning – from ECDIS in HiNAS Control Standard 3. Set departure time as now, and arrival time as certain time 4. Calculate voyage optimization 5. Check if the latitude and longitude coordinates of the Applied Route from the EUT match those of the Route data from the ECDIS		v	


HiNAS Control
Route Mismatch
ACK
Route Info
WPT
Function Act
Autopilot
Standalone
Track Control
CA Assistance
CA Control
Ship Status
Environment
WPT6
From ECDIS
Name 62065_sen4
Reverse
Departure
Arrival
42°22.38' N
070°54.21' W
51°58.86' N
004°04.61' E
2025-01-17 10:59
Loading Condition
Margin
Route
Waypoint
XTD [NM]
No.
LAT
LON
Route Type
Turn Radius [NM]
PORT
STBD
0 42°22.38' N 070°54.21' W
1 42°23.27' N 070°47.66' W
2 42°27.29' N 070°35.95' W
RL
GC
Cancel
Calculate
TGT Monitoring DIST 20 NM
Version 25.0.0
WGS 84 | Mercator | Non-ECDIS Presentation


1.
2.
3.
4.
Lost Network
ACK
Previous Routes
Departure
Arrival
Loading Condition
Margin
Route
Cancel
Calculate


Lost Network
ACK
Validate
1.
2.
Apply
Export
Back

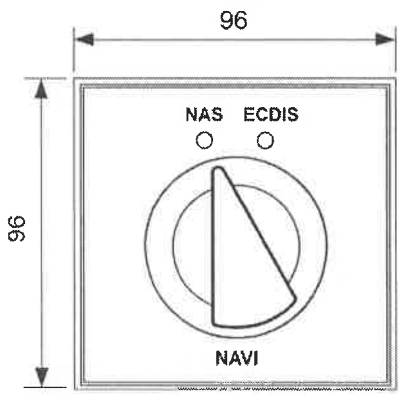
<6. Confirm Applied Route >

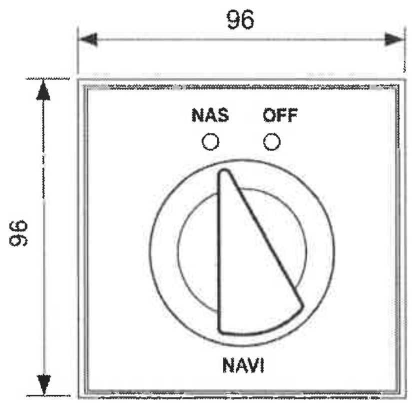
Comment / Recommended

3.4 Autopilot fail test

Relevant Standard	Standard No. IEC62065		
	Article. 5.1.1.14 and 5.5.1		
Purpose	The purpose of this test is to evaluate alarm status information performance test when certain interface is disconnected and to check the alarm level is correspondence with design requirement		
Procedure of test		Pass	Fail
1. Ensure that the Autopilot firmware version meets the specifications defined in MED-B before proceeding with the test. 2. Prepare HiNAS Control Standard voyage planning and autopilot as normal condition, ensure autopilot will not affect the rudder. 3. Press TC button of HiNAS Control Standard 4. Change manual steering mode to heading control mode of autopilot 5. (TKM) Press Ext.HC of Autopilot 6. Disconnect autopilot by - Turning "mode selection unit" to "ECDIS" or "OFF" - If the ECDIS equipment is equipped with a Track Control function, indicate “ECDIS” on the switch when the NAS device is disconnected.		v	

- If the ECDIS equipment is not equipped with a Track Control function, indicate "OFF" on the switch when the NAS device is disconnected.





5. Verify the following:
- 'Track control stopped' or its equivalent alert is raised in alert list with buzzer sound.

3007	1	TRK Control Stop	Last command will be maintained. Take over the control.	Alarm / B	NA	TC

* Note: Alert list page may vary with ship condition

Comment / Recommended

3.5 Sensor interface fail test

Relevant Standard	Standard No. IEC62065
	Article. 5.1.3.2, 5.1.3.3, 5.5.2, 5.5.3, 5.5.4
Purpose	The purpose of this test is to evaluate alarm status information performance test when certain interface is disconnected and to check the alarm level is correspondence with design requirement

Procedure of test	Pass	Fail
1. Same procedure of 1~4 of 3.4. 2. Disconnect all connection to sensors 3. Verify following - 'Lost Network' or its equivalent alert is raised in alert list 	v	

3001	-	Lost Network	All function will stop. Take over the control. Check the onboard network status connected to HiNAS Control.	Alarm / B	NA	TC, CAC

* Note: Alert list page may vary with ship condition

4. Same procedure of 1
5. Disconnect position sensor
6. Verify following

- 'Lost Position' or its equivalent alert is raised in alert list

3014	102	Lost SOG/COG	Track Control performance can be reduced. Change the primary SOG/COG sensor, or verify its status and connection to recover the performance of Track Control.	Alarm / B	NA	TC

7. Same procedure of 1
8. Disconnect heading sensor
9. Verify following

- 'Lost Heading' or its equivalent alert is raised in alert list

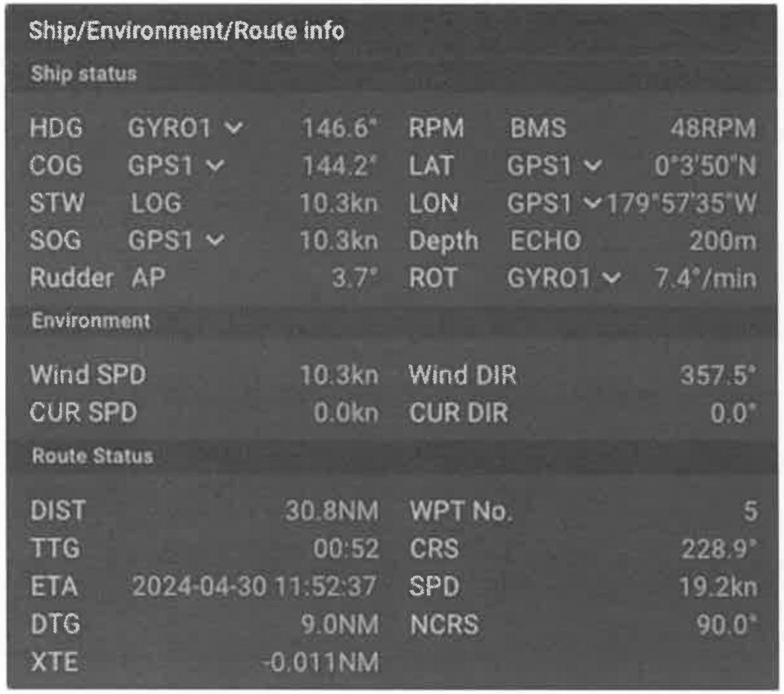
3014	104	Lost Heading	Track Control will stop. Take over control. Change the primary heading sensor, or verify its status and connection to restore Track Control.	Alarm / B	NA	TC

10. Same procedure of 1
11. Disconnect speed sensor
12. Verify following

- 'Lost STW' or its equivalent alert is raised in alert list

3014	106	Lost STW	Track Control performance can be reduced. Change the primary STW sensor, or verify its status and connection to recover the performance of Track Control.	Alarm / B	NA	TC

* Note: Signal lost can be found in below status display

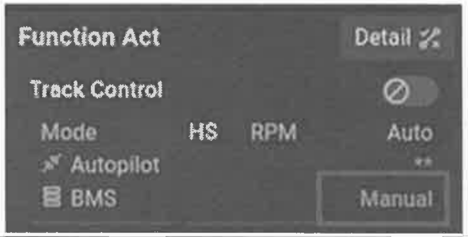


Comment / Recommended

3.6 BMS Fail Test

Relevant Standard	
Purpose	The purpose of this test is to evaluate alarm status information performance test when certain interface is disconnected and to check the alarm level is correspondence with design requirement

Procedure of test	Pass	Fail
1. Ensure BMS is turned on 2. Disconnect BMS interface cable 3. Verify following - BMS status in HiNAS Control Standard is changed to "**"	v	



Comment / Recommended

3.7 Power Fail Test

Relevant Standard	Standard No. IEC62065
	Article 5.1.3.1.
Purpose	The purpose of this test is to evaluate alarm status information performance test when certain interface is disconnected and to check the alarm level is correspondence with design requirement

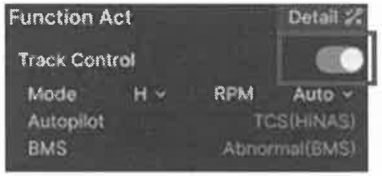
Procedure of test	Pass	Fail
1. Remove power from the UPS of HiNAS Control Standard 2. Verify following - 'Power fail' or its equivalent alert is raised in alert list 3022 | - | Power fail | HiNAS Control detected power failure. Take over the control. Check the power source. | Warning / B | Repeat Warning | - - 'Lan fail' alert is raised in alert list 777109 | - | Lost UPS | HiNAS Control cannot detect the power failure. Check the network connection between HiNAS Control and UPS. | Caution / B | - | -	V	

Comment / Recommended

3.8 Operation Mode Change Test – Mode Selection Unit

Relevant Standard	Standard No.		
	Article		
Purpose	The purpose of this test is to test auto pilot mode is changed to stand-alone mode after HiNAS mode change		
Procedure of test		Pass	Fail
1. Mode change by mode selection unit. - Common for test item 3.4		v	
Comment / Recommended			

3.9 Operation Mode Change Test – SW Button

<table>
<tr><td rowspan="2">Relevant Standard</td><td colspan="3">Standard No.</td></tr>
<tr><td colspan="3">Article</td></tr>
<tr><td>Purpose</td><td colspan="3">The purpose of this test is to test auto pilot mode is changed to stand-alone mode after HiNAS mode change</td></tr>
<tr><td colspan="2">Procedure of test</td><td>Pass</td><td>Fail</td></tr>
<tr><td colspan="2">1. Same procedure of 1~4 of 3.4.
2. Deactivate Track Control function of HiNAS (SW button)

3. Verify following
- Operation mode of Autopilot is changed to HC(or equivalent mode) mode</td><td>v</td><td></td></tr>
<tr><td colspan="4">Comment / Recommended</td></tr>
<tr><td colspan="4"></td></tr>
</table>